

JOHN NILO AVANCEÑA

AI Automation Engineer · Software Engineer · Full-Stack Developer

Remote (Global) · Metro Manila, Philippines · Open to remote & relocation

+63 921 457 1796 · avancena.johnilo@gmail.com

linkedin.com/in/john-nilo-avanceña-0777b7242 · github.com/your-username

PROFESSIONAL SUMMARY

Software engineer specializing in AI-powered automation and LLM integration. Delivered contract work for SourceOne Global replacing manual operations data entry with a Claude-powered document pipeline synced to NetSuite and Monday.com, and builds production systems that turn unstructured input — PDFs, resume emails, livestream video — into structured output using the Anthropic Claude, Google Gemini, and Ollama APIs. Experienced across Next.js, Python (FastAPI), and ASP.NET Core with real-time systems, background job queues, vector embeddings, multi-tenant data isolation, OAuth/JWT auth, and deployment on Azure and Vercel. Enterprise foundation in ERP customization (SAP ABAP, Microsoft Dynamics AX).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, X++, ABAP

AI / LLM: Anthropic Claude API, Google Gemini API, Ollama, Whisper (faster-whisper), OpenCV, Prompt Engineering, Structured / JSON-Schema Output, Vector Embeddings (pgvector), Semantic Search, Prompt Caching, Multi-Provider LLM Abstraction

Frameworks: Next.js, React 19, React Native (Expo), Node.js, FastAPI, ASP.NET Core (.NET 8), SignalR, .NET MAUI, Flask, Tailwind CSS

Automation: Document/PDF Extraction Pipelines, Gmail API ingestion, Background Job Queues, Scheduled Cron Workers, Event-Driven Notifications, n8n Webhooks, FFmpeg, SMTP

APIs: REST, GraphQL (Monday.com), NetSuite SuiteTalk / SuiteQL, WebSockets (SignalR), Gmail API, OAuth 2.0 (incl. M2M JWT assertions), Google Maps, Swagger/OpenAPI

Databases: PostgreSQL (Supabase, pgvector), MongoDB (2dsphere geospatial), SQL Server

Cloud & DevOps: Azure App Service, Vercel, Supabase, Docker, Git, GitHub, CI/CD

Security & Auth: JWT, OAuth 2.0, Row-Level Security, RBAC, Multi-Tenancy, BCrypt, Rate Limiting, CORS

PROJECTS

AI Recruitment Automation Platform — Multi-Tenant SaaS

Next.js 16 · React 19 · TypeScript · Supabase (PostgreSQL / RLS) · Claude · Gemini · Ollama · Gmail API · OAuth 2.0 · Vercel Cron · n8n

- Architected a multi-tenant AI recruitment SaaS that ingests resume emails and returns AI-screened, ranked candidates with no manual data entry, enforcing tenant isolation through PostgreSQL Row-Level Security and RBAC (admin / recruiter / interviewer).
- Built a provider-agnostic LLM abstraction exposing one structured-output interface over the Anthropic Claude, Google Gemini, and Ollama APIs, using JSON-schema-constrained generation and native PDF input to parse resumes and score candidate–job fit reliably.
- Engineered an asynchronous background evaluation queue with compare-and-swap job claiming, automatic stuck-row crash recovery, and batch draining via Next.js `after()` and an authenticated Vercel Cron endpoint — keeping the LLM off the request hot path.
- Implemented an end-to-end Gmail OAuth ingestion pipeline that routes applications by deterministic email-subject matching (never the LLM), stores resume PDFs in object storage, auto-replies on format errors, deduplicates by message ID, and logs every decision.
- Designed approval-gated automation endpoints (n8n webhook orchestration) so candidate-facing emails and pipeline stage changes execute only after human approval — cleanly separating AI parsing/scoring from irreversible actions.

AI Video Clipping Pipeline — Speech-to-Text + LLM + Computer Vision

Python · faster-whisper (CUDA) · Google Gemini · FFmpeg (NVENC) · OpenCV · Flask

- Built an end-to-end AI video-automation pipeline that converts long livestream VODs into captioned vertical short-form clips, orchestrating six stages from download to final render behind one command and a Flask GUI.
- Integrated GPU-accelerated speech-to-text (word-level timestamps) with graceful CUDA→CPU fallback — dynamically registering cuDNN/cuBLAS libraries and downshifting model size — so the pipeline runs on constrained hardware without failing.
- Applied prompt engineering and schema-guided LLM output to select hook-forward clip segments, hardening the parser with retry, truncated-JSON salvage, word-boundary snapping, and overlap de-duplication.

- Automated rendering with FFmpeg (NVENC 9:16 reframing, burned-in captions) and OpenCV YuNet face detection that selects per-clip framing with a guaranteed center-crop fallback so detection errors never abort a run.

Real-Time PUV Fleet Tracking Platform

ASP.NET Core (.NET 8) · SignalR · MongoDB (2dsphere) · JWT · React Native (Expo) · React + Leaflet · Azure App Service · Docker

- Engineered a real-time fleet-tracking platform that streams live vehicle positions to commuter/driver mobile apps and an operator web console, using route-scoped WebSocket groups to fan out only relevant updates as it scales.
- Designed a MongoDB geospatial model (GeoJSON + 2dsphere index) serving “vehicles near me” radius queries via \$nearSphere with no collection scan.
- Implemented multi-tenant isolation with a per-request tenant context resolved from JWT claims and company-scoped repositories, plus an additive platform-admin authorization policy.
- Secured the API with JWT bearer auth (incl. a WebSocket query-string token handoff for SignalR), per-IP rate limiting on auth endpoints, BCrypt hashing, strict CORS, and KYC ID/selfie verification at registration.
- Built resilient background services (auto-closing abandoned driver shifts; telemetry sampling) and a single authoritative write path with REST reconciliation for a self-healing live map.

EXPERIENCE

Software Engineer (Contract)

06/2026

SourceOne Global

- Automated the operations team's manual workflow — the same order data was being typed into NetSuite, Monday.com, and Google Sheets — with a drag-and-drop portal: upload a quotation PDF and structured sales-order data flows to every downstream system.
- Integrated the Anthropic Claude API for native-PDF field extraction with a strict extraction prompt; extracted data prefills a human-review form and is never persisted without operator confirmation.
- Implemented NetSuite integration over SuiteTalk REST with OAuth 2.0 machine-to-machine auth — PS256-signed JWT client assertions with token caching — and SuiteQL sales-order reads.
- Automated Monday.com workflow boards via the GraphQL API: project-grouped item creation, typed column updates, and attaching the source quotation PDF to the board.
- Built a semantic product-matching pipeline combining pgvector embedding retrieval with a Claude Sonnet final-selection stage — prompt caching to cut repeat-call cost, versioned prompts persisted per match for auditability — inside a FastAPI + Next.js + Supabase portal.

SAP ABAP Developer — Academy Delegate

02/2026 – 06/2026

Accenture Philippines

- Developed Object-Oriented ABAP programs, reports, and classes using SAP GUI / SAP Logon for modular, reusable execution.
- Performed structured debugging with the ABAP Debugger — breakpoints, stack analysis, and runtime navigation across includes, events, and screen stacks.
- Worked across core SAP development transactions (SE80, SE24, SE11, SE37, SE16N, SCI, SE30).

Microsoft Dynamics AX 2012 Developer

03/2025 – 01/2026

Creative Dynamics Solutions Inc.

- Designed and customized forms and UI with X++ to support business workflows; implemented business logic via X++ controllers, data providers, and services.
- Built SSRS reports in Visual Studio with multiple datasets and custom AX views/tables; automated financial document generation (AP vouchers, check vouchers, sales invoices).
- Debugged and resolved issues across AX modules, improving performance and functionality; delivered customizations for clients including LHPI and Cirtek Electronics Corporation.
- Collaborated with functional consultants to translate business requirements into AX customizations.

EDUCATION

Bachelor of Science in Information Technology

STI College Cubao

CERTIFICATIONS / TRAINING

Accenture Academy — SAP ABAP Development & SAP End-to-End Basis